hydrogen, carbon, oxygen, nitrogen, calcium, iron, and so forth. But more important than our laundry list of shared ingredients was the recognition that the laws of physics prescribing the formation of these spectral signatures on the Sun were the same laws operating on Earth, ninety-three million miles away.

So fertile was this concept of universality that it was successfully applied in reverse. Further analysis of the Sun's spectrum revealed the signature of an element that had no known counterpart on Earth. Being of the Sun, the new substance was given a name derived from the Greek word *helios* ("the Sun"), and was only later discovered in the lab. Thus, helium became the first and only element in the chemist's Periodic Table to be discovered someplace other than Earth.

Okay, the laws of physics work in the solar system, but do they work across the galaxy? Across the universe? Across time itself? Step by step, the laws were tested. Nearby stars also revealed familiar chemicals. Distant binary stars, bound in mutual orbit, seem to know all about Newton's laws of gravity. For the same reason, so do binary galaxies.

And, like the geologist's stratified sediments, which serve as a timeline of earthly events, the farther away we look in space, the further back in time we see. Spectra from the most distant objects in the universe show the same chemical signatures that we see nearby in space and in time. True, heavy elements were less abundant back then—they are manufactured primarily in subsequent generations of exploding stars—but the laws describing the atomic and molecular processes that created these spectral signatures remain intact. In particular, a quantity known as the fine-structure constant, which controls the basic fingerprinting for every element, must have remained unchanged for billions of years.

Of course, not all things and phenomena in the cosmos have counterparts on Earth. You've probably never walked through a cloud of glowing million-degree plasma, and I'd bet you've never greeted a black hole on the street. What matters is the universality of the physical laws that describe them. When spectral analysis was first applied to the light emitted by interstellar nebulae, a signature was discovered that, once again, had no counterpart on Earth. At the time, the Periodic Table of Elements had no obvious place for a new element to fit. In response, astrophysicists invented the name "nebulium" as a place-holder, until they could figure out what was going on. Turned out that in space, gaseous nebulae are so rarefied that atoms go long stretches without colliding. Under these conditions, electrons can do things within atoms